BIM488 Introduction to Pattern Recognition

Text Classification

Outline

- Introduction
- Text representation
- Text classification
- Term Selection

Introduction

- Text classification/categorization is a problem in information science.
- The task is to assign a document to one or more categories, based on its contents.

In other words:

- given a predefined set of categories and a set of documents
- label each document with one or more categories

Text representation

- selection of terms
- vector model
- weighting (TD*IDF)

Text representation

- text cannot be directly interpreted by the many document processing applications
- we need a compact representation of the content
- which are the meaningful units of text?

Terms

- Words
  - typical choice
  - set of words, bag of words
- Phrases
  - syntactical phrases (e.g. noun phrases)
  - statistical phrases (e.g. frequent pairs of words)
  - usefulness not yet known?

Terms

- Stop-word removal: part of the text is not considered as terms: these words can be removed
  - very common words (function words):
    - articles (a, the) , prepositions (of, in), conjunctions (and, or), adverbs (here, then)
  - numerals (30.9.2002, 2547)
- other preprocessing steps
  - Stemming (i.e., apples → apple)

Vector model

- a document is often represented as a vector
- the vector has as many dimensions as there are terms in the whole collection of documents

Vector model

- Assume in a sample document collection, there are 100 words (terms)
- In alphabetical order, the list of terms starts with:
  - absorption
  - agriculture
  - anaemia
  - analyse
  - application
  - ...

Vector model

- Each document can be represented by a vector of 100 dimensions
- We can think a document vector as an array of 100 elements, one for each term, indexed, e.g. 0-99

Vector model

- let d1 be the vector for document 1
- record only which terms occur in document:
  - d1[0] = 0 -- absorption doesn't occur
  - d1[1] = 0 -- agriculture -"-
  - d1[2] = 0 -- anaemia -"-
  - d1[3] = 0 -- analyse -"-
  - d1[4] = 1 -- application occurs
  - ...
  - d1[21] = 1 -- current occurs
  - ...

Weighting terms

- usually we want to say that some terms are more important (for some document) than the others -> weighting
- weights usually range between 0 and 1
  - 1 denotes presence, 0 absence of the term in the document

Weighting terms

- if a word occurs many times in a document, it may be more important
  - but what about very frequent words?
- often the **TF*IDF** function is used
  - higher weight, if the term occurs often in the document
  - lower weight, if the term occurs in many documents

Weighting terms: TF*IDF

- TF*IDF = term frequency * inversed document frequency
- weight of term t_k in document d_j:

$$tfidf(t_k, d_j) = \#(t_k, d_j) \cdot \log \frac{|Tr|}{\#Tr(t_k)}$$

- where
  - $\#(t_k,d_j)$: the number of times t_k occurs in d_j
  - $\#Tr(t_k)$: the number of documents in Tr in which t_k occurs
  - Tr: the documents in the collection

Weighting terms: TF*IDF

- in document 1:
  - term ’application’ occurs once, and in the whole collection it occurs in 2 documents:
    - tfidf (application, d1) = 1 * log(10/2) = log 5 ~ 0.7
  - term ´current´ occurs once, in the whole collection in 9 documents:
    - tfidf(current, d1) = 1 * log(10/9) ~ 0.05

Weighting terms: TF*IDF

- if there were some word that occurs 7 times in doc 1 and only in doc 1, the TF*IDF weight would be:
  - tfidf(doc1word, d1) = 7 * log(10/1) = 7

Weighting terms: normalization

- in order for the weights to fall in the [0,1] interval, the weights are often normalized (T is the set of terms):

$$w_{kj} = \frac{tfidf(t_k, d_j)}{\sqrt{\sum_{s=1}^{|T|} (tfidf(t_s, d_j))^2}}$$

Text categorization

- two major approaches:
  - knowledge engineering -> end of 80's
    - manually defined set of rules encoding expert knowledge on how to classify documents under the given gategories
  - machine learning, 90's ->
    - an automatic text classifier is built by learning, from a set of preclassified documents, the characteristics of the categories

Single-label, multi-label TC

- single-label text categorization
  - exactly 1 category must be assigned to each $d_j \in D$
- multi-label text categorization
  - any number of categories may be assigned to the same $d_j \in D$

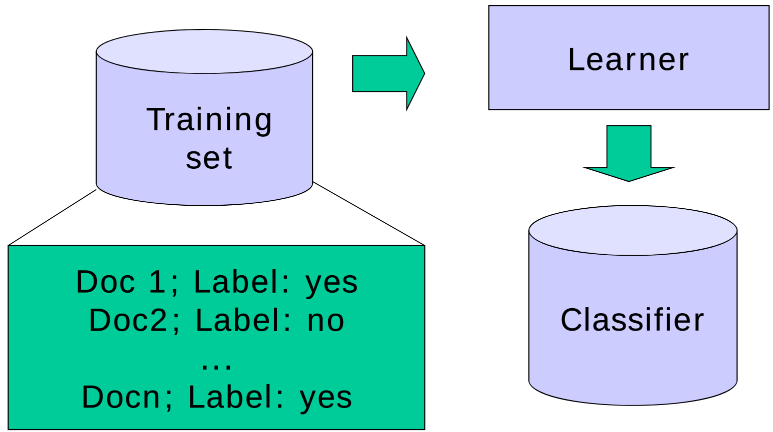
Single-label, multi-label TC

- special case of single-label: binary
  - each d_j must be assigned either to category c_i or to its complement $\neg c_i$
- the binary case (and, hence, the single-label case) is more general than the multi-label
  - an algorithm for binary classification can also be used for multi-label classification
  - the converse is not true

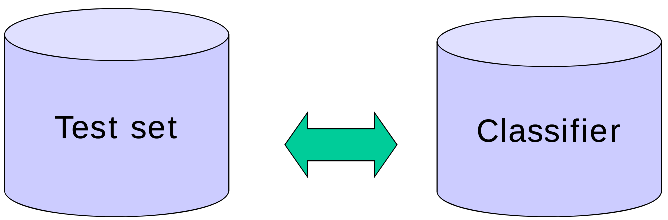
Machine learning approach

- a general inductive process (learner) automatically builds a classifier for a category c_i by observing the characteristics of a set of documents manually classified under c_i or $\neg c_i$ by a domain expert
- from these characteristics the learner extracts the characteristics that a new unseen document should have in order to be classified under c_i
- use of classifier: the classifier observes the characteristics of a new document and decides whether it should be classified under c_i or $\neg c_i$

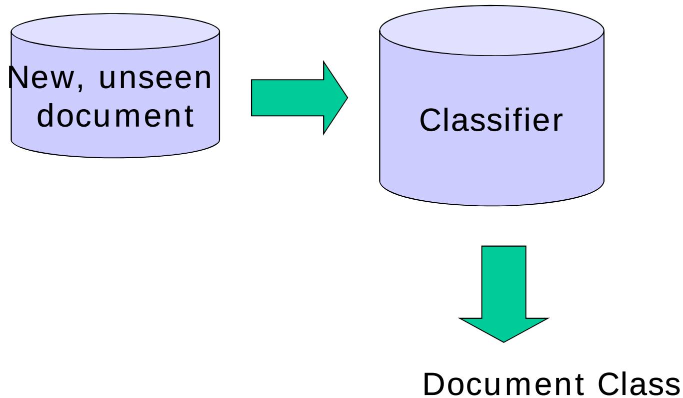
Classification process: classifier construction



Classification process: testing the classifier



Classification process: use of the classifier



Strengths of machine learning approach

- the learner is domain independent
  - usually available 'off-the-shelf'
- the inductive process is easily repeated, if the set of categories changes
  - only the training set has to be replaced
- manually classified documents often already available
  - manual process may exist
  - if not, it is still easier to manually classify a set of documents than to build and tune a set of rules

Examples of learners

- Rocchio method
- probabilistic classifiers (Naïve Bayes)
- decision tree classifiers
- decision rule classifiers
- regression methods
- on-line methods
- neural networks
- example-based classifiers (k-NN)
- boosting methods
- support vector machines

Term selection

- a large document collection may contain millions of words -> document vectors would contain millions of dimensions
  - many algorithms cannot handle high dimensionality of the term space (= large number of terms)
  - very specific terms may lead to overfitting: the classifier can classify the documents in the training data well but fails often with unseen documents

Term selection

- usually only a part of terms is used
- how to select terms that are used?
  - term selection (often called feature selection or dimensionality reduction) methods

Term selection

- goal: select terms that yield the highest effectiveness in the given application
- wrapper approach
  - the reduced set of terms is found iteratively and tested with the application
- filtering approach
  - keep the terms that receive the highest score according to a function that measures the ”importance” of the term for the task

Term selection

- many functions available
  - document frequency: keep the high frequency terms
    - stopwords have been already removed
    - 50% of the words occur only once in the document collection
    - e.g. remove all terms occurring in at most 3 documents

Term selection functions: document frequency

- document frequency is the number of documents in which a term occurs
- in our sample, the ranking of terms:
  - 9 current
  - 7 project
  - 4 environment
  - 3 nuclear
  - 2 application
  - 2 area … 2 water
  - 1 use …

Term selection functions: document frequency

- we might now set the threshold to 2 and remove all the words that occur only once
- result: 25 words of 100 words (~25%) selected

Term selection: other functions

- Information-theoretic term selection functions, e.g.
  - chi-square
  - information gain
  - mutual information
  - odds ratio
  - relevancy score

Term selection: information gain

- Information gain: measures the (number of bits of) information obtained for category prediction by knowing the presence or absence of a term in a document
- information gain is calculated for each term and the best n terms are selected

Term selection: IG

- information gain for term t:
  - m: the number of categories

$$G(t) = -\sum_{i=1}^{m} p(c_i)\log p(c_i)$$
$$+p(t)\sum_{i=1}^{m} p(c_i \mid t)\log p(c_i \mid t)$$
$$+p(\sim t)\sum_{i=1}^{m} p(c_i \mid \sim t)\log p(c_i \mid \sim t)$$

Estimating probabilities

2 classes: c and ~c

- (c) Doc 1: cat cat cat
- (c) Doc 2: cat cat cat dog
- (~c) Doc 3: cat dog mouse
- (~c) Doc 4: cat cat cat dog dog dog
- (~c) Doc 5: mouse

Term selection: estimating probabilities

- P(t): probability of a term t
  - P(cat) = 4/5, or
    - ‘cat’ occurs in 4 docs of 5
  - P(cat) = 10/17
    - the proportion of the occurrences of ´cat’ of the all term occurrences

Term selection: estimating probabilities

- P(~t): probability of the absence of t
  - P(~cat) = 1/5, or
  - P(~cat) = 7/17

Term selection: estimating probabilities

- $P(c_i)$: probability of category i
  - P(c) = 2/5 (the proportion of documents belonging to c in the collection), or
  - P(c) = 7/17 (7 of the 17 terms occur in the documents belonging to c)

Term selection: estimating probabilities

- $P(c_i \mid t)$: probability of category i if t is in the document; i.e., which proportion of the documents where t occurs belong to the category i
  - $P(c \mid cat) = 2/4$ (or 6/10)
  - $P(\sim c \mid cat) = 2/4$ (or 4/10)
  - $P(c \mid mouse) = 0$
  - $P(\sim c \mid mouse) = 1$

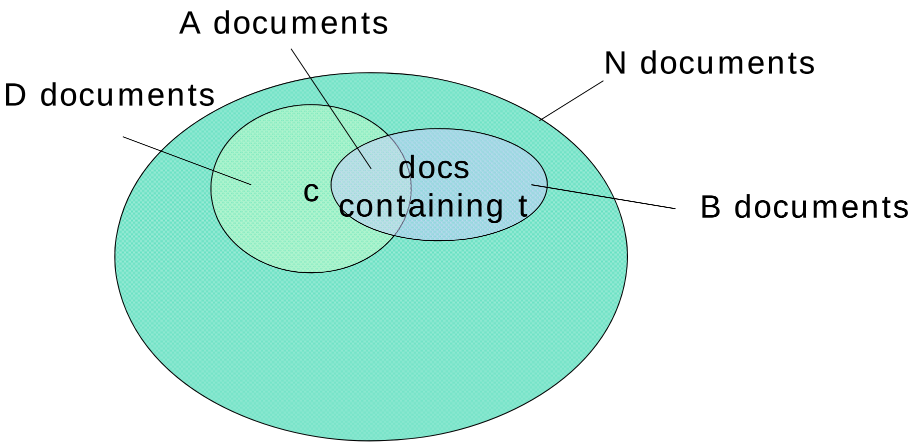
Term selection: estimating probabilities

- $P(c_i \mid \sim t)$: probability of category i if t is not in the document; i.e., which proportion of the documents where t does not occur belongs to the category i
  - $P(c \mid \sim cat) = 0$ (or 1/7)
  - $P(c \mid \sim dog) = ½$ (or 6/12)
  - $P(c \mid \sim mouse) = 2/3$ (or 7/15)

Term selection: estimating probabilities

- In other words...
- Let
  - term t occurs in B documents, A of them are in category c
  - category c has D documents, of the whole of N documents in the collection

Term selection: estimating probabilities



Term selection: estimating probabilities

- For instance,
  - $P(t)$: B/N
  - $P(\sim t)$: $(N-B)/N$
  - $P(c)$: D/N
  - $P(c|t)$: A/B
  - $P(c|\sim t)$: $(D-A)/(N-B)$

Term selection: IG

- information gain for a term t:

$$G(t) = -\sum_{i=1}^{m} p(c_i)\log p(c_i)$$
$$+p(t)\sum_{i=1}^{m} p(c_i \mid t)\log p(c_i \mid t) + p(\sim t)\sum_{i=1}^{m} p(c_i \mid \sim t)\log p(c_i \mid \sim t)$$

- G(cat) = 0.17
- G(dog) = 0.02
- G(mouse) = 0.42

Summary

- Introduction
- Text representation
- Text classification
- Term Selection

References

- 582410 Processing of large document collections, Lecture Notes, University of Helsinki.